A Closed-Form, Non-Recursive Move Formula for the Tower of Hanoi, and Its Behavior on Non-Canonical Initial Configurations

Abstract

The Tower of Hanoi is almost universally solved by a two-line recursive algorithm. We revisit a closed-form alternative: a formula that, given only a move index m, the disk count n, and a source–destination peg pair (s, f), returns the disk moved on turn m and the peg pair it travels between — with no recursion, no stack, and no simulated state. We give a self-contained derivation of this formula from first principles, showing it is equivalent to two classical facts about the optimal Hanoi sequence: the *ruler sequence* that governs which disk moves on turn m (OEIS A001511), and the fact that the smallest disk cycles around the three pegs in a single fixed rotational direction determined by the parity of n and by (s, f). We then examine an extension used by the reference implementation to handle *arbitrary* legal starting configurations (disks distributed arbitrarily over the three pegs, not merely stacked on one), by applying the closed-form formula disk-by-disk from the inside out. We prove this construction always terminates in a valid sequence of legal moves, but we show empirically that it is frequently and sometimes severely suboptimal on scattered configurations: in a sample of 300 random legal 3–5-disk configurations, roughly one third were solved with strictly more moves than the true graph-distance optimum, with one case using 51 moves against an optimal value of 24. We give an exact formula for its (generally supra-optimal) move count and relate the resulting gap to the long-standing open Frame–Stewart-type multi-peg literature, of which the scattered three-peg case is a close cousin.

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1 Introduction

The Tower of Hanoi puzzle, introduced by Édouard Lucas in 1883 under the pseudonym "N. Claus de Siam" [1], asks for the minimum number of moves required to transfer a tower of n disks of distinct sizes, initially stacked in decreasing order of size on one of three pegs, to another peg, subject to the rule that a disk may never be placed on top of a smaller one. The minimum number of moves is classically known to be

$$M(n) = 2^n - 1,$$

achieved by the familiar recursive strategy: move the top $n-1$ disks out of the way, move the largest disk, then move the $n-1$ disks back on top of it. This recursive algorithm, and the resulting binary structure of the optimal move sequence, has been studied extensively; see Hinz, Klavžar, Milutinović, and Petr [2] for a comprehensive modern treatment, including the connection between the optimal move sequence and the *ruler sequence* $\nu_2(m) + 1$ (the 2-adic valuation of m, plus one), and the classical fact — attributed to the "binary" or non-recursive solution methods described already in the nineteenth century [1, 2] — that the smallest disk moves in a single fixed rotational direction around the pegs on every one of its turns.

The present paper documents and formally justifies a small piece of software that computes the optimal move sequence *without* recursion: a single closed-form arithmetic expression, evaluated independently for each move index $m = 1, \ldots, 2^n - 1$, that returns which disk moves and between which two pegs. Non-recursive Hanoi algorithms of this general flavor are not new — closed-form and bit-trick solutions appear in recreational and pedagogical sources going back decades, and the underlying number-theoretic facts (the ruler sequence, and the fixed-direction cycling of each disk) are standard [2, 5]. Our contributions here are: (i) a complete, checked, from-scratch derivation of the specific closed-form expression used by the implementation, expressed in the exact bit-level and modular-arithmetic form in which it appears in the code, with an explicit proof of correctness rather than a citation to folklore; and (ii) an analysis — both exact and empirical — of what happens when this formula's disk-by-disk construction is applied to configurations that do not start as a single tower on one peg, a case the classical $2^n - 1$ theory does not directly cover and for which we show the construction is correct but generically non-optimal, quantifying the gap on random instances.

1.1 Overview

Section 2 fixes notation and states the classical results we build on. Section 3 states the closed-form transition formula exactly as implemented, and proves it correct for the canonical (single-peg start) case. Section 4 extends the analysis to arbitrary legal starting configurations, describes the disk-by-disk ("layered") construction used to handle them, proves it always produces a valid solution, gives an exact formula for the number of moves it uses, and reports an empirical study of its optimality gap against the true shortest-path distance, computed by breadth-first search on the configuration graph. Section 5 discusses time and space complexity. Section 7 concludes.

2 Setup and Classical Background

Definition 1 (Configuration)**.** *Fix $n \geq 1$ disks, labeled $1, 2, \ldots, n$ in increasing order of size, and three pegs labeled $1, 2, 3$. A* configuration *is a function $\rho : \{1, \ldots, n\} \to \{1, 2, 3\}$ assigning each disk to a peg, such that whenever $\rho(i) = \rho(j)$ for $i < j$, disk i (smaller) is above disk j (larger) in the stacking order on that peg. Equivalently, a configuration is an assignment of disks to pegs together with, for each peg, the unique consistent (size-decreasing bottom-to-top) stacking of the disks assigned to it. A configuration is* legal *if it satisfies this size constraint on every peg; we consider only legal configurations throughout.*

Definition 2 (Move, move sequence)**.** *A* move *takes the topmost disk on some peg x and places it on peg $y \neq x$, provided the result is legal (i.e. peg y is empty or its current top disk is larger). A* move sequence *solving (ρ_0, ρ_1) is a finite sequence of moves transforming configuration ρ_0 into configuration ρ_1.*

Definition 3 (Canonical configuration)**.** *A configuration is* canonical *with source s if $\rho(i) = s$ for all i, i.e. all n disks are stacked on a single peg s.*

The classical Tower of Hanoi problem asks for a shortest move sequence between two canonical configurations with distinct pegs $s \neq f$. We will call this the *canonical problem* $H(n, s, f)$.

Theorem 4 (Lucas, 1883; see [1, 2])**.** *For $n \geq 1$ and $s \neq f$, the canonical problem $H(n, s, f)$ has a unique shortest move sequence (up to the trivial symmetry of relabeling the third, unused peg), of length*

$$M(n) = 2^n - 1.$$

We will re-derive the exact move sequence achieving Theorem 4 in closed form; uniqueness and the value $2^n - 1$ we take as given (both are elementary consequences of the standard recursive argument, and are not the focus of this paper).

2.1 The ruler sequence

Definition 5 (2-adic valuation)**.** *For a positive integer m, let $\nu(m)$ denote the exponent of the largest power of 2 dividing m, i.e. $m = 2^{\nu(m)} \cdot (\text{odd number})$.*

The sequence $\nu(1), \nu(2), \nu(3), \cdots = 0, 1, 0, 2, 0, 1, 0, 3, \ldots$ is the *ruler sequence* (OEIS A007814); the shifted sequence $\nu(m) + 1$ is OEIS A001511 and is classically known to describe which disk moves on turn m of the optimal Hanoi solution [2].

Lemma 6. *In the unique shortest move sequence solving $H(n, s, f)$, the disk moved on turn m (for $1 \leq m \leq 2^n - 1$) is disk*

$$r(m) = \nu(m) + 1.$$

Proof. By induction on n. For $n = 1$ the sequence has one move, moving disk 1, and $r(1) = \nu(1) + 1 = 0 + 1 = 1$, as required. For the inductive step, the standard recursive solution to $H(n, s, f)$ consists of: the (unique, by induction) solution to $H(n-1, s, t)$ where t is the third peg (moves 1 through $2^{n-1} - 1$), then a single move of disk n (move 2^{n-1}), then the solution to $H(n-1, t, f)$ (moves $2^{n-1} + 1$ through $2^n - 1$). For $1 \leq m \leq 2^{n-1} - 1$, m is unchanged from the sub-instance and $\nu(m) + 1 \leq n - 1 < n$, matching the disk moved in $H(n-1, s, t)$ by the inductive hypothesis. At $m = 2^{n-1}$, $\nu(2^{n-1}) = n - 1$, so $r(m) = n$, matching the single move of the largest disk. For $2^{n-1} < m \leq 2^n - 1$, write $m = 2^{n-1} + m'$ with $1 \leq m' \leq 2^{n-1} - 1$; since $m' < 2^{n-1}$, $\nu(m) = \nu(m')$ (adding 2^{n-1} does not change the number of trailing binary zeros of a number strictly smaller than 2^{n-1}), so $r(m) = \nu(m') + 1$, matching the disk moved at position m' of $H(n-1, t, f)$ by the inductive hypothesis. □

2.2 Fixed-direction cycling

Lemma 7. *In the unique shortest move sequence solving $H(n, s, f)$, each disk k moves, on each of its 2^{n-k} turns, in the same fixed rotational sense around the cycle $1 \to 2 \to 3 \to 1$ or $1 \to 3 \to 2 \to 1$; disks of opposite parity move in opposite senses, and the sense of disk n (hence, by parity, every disk) is determined by the parity of n and by the pair (s, f).*

This is again classical (see [2], and the "rule of thumb" non-recursive Hanoi strategies described there); we give a direct combinatorial argument as part of the proof of Theorem 11 below, since it is precisely the fact that the closed-form formula must encode.

3 The Closed-Form Transition Formula

We now state the formula exactly as implemented, using the disk-index convention above (disk 1 = smallest), and prove it correct.

Definition 8 (Transition formula)**.** *For $n \geq 1$, pegs $s \neq f \in \{1, 2, 3\}$, and $1 \leq m \leq 2^n - 1$, define*

$$r(m) = \nu(m) + 1, \tag{1}$$

$$d(n, s, f) = (-1)^{n \bmod 2 + (f-s) \bmod 3}, \tag{2}$$

$$k(m) = \frac{m}{2^{r(m)-1}} - \frac{1}{2}, \tag{3}$$

$$x(m) = 1 + \Big(s - 1 + d(n, s, f)\, k(m) \big(2 - (r(m) \bmod 2)\big)\Big) \bmod 3, \tag{4}$$

$$y(m) = 1 + \Big(x(m) - 1 + d(n, s, f) \big(2 - (r(m) \bmod 2)\big)\Big) \bmod 3. \tag{5}$$

Here mod *denotes the representative in $\{0, 1, 2\}$ (equivalently, in $\{0, \ldots, N-1\}$ for modulus N). Move m of $H(n, s, f)$ is then declared to move disk $r(m)$ from peg $x(m)$ to peg $y(m)$.*

Equation (1)–Equation (5) are a direct transcription of the function `compute_move_transition(n, s, f, m)` in the reference implementation, where $\nu(m) + 1$ is computed via the bit-trick `((2*m) & -(2*m)).bit_length() - 1` (the position of the lowest set bit of $2m$, using two's-complement negation to isolate it), which is a standard constant-time idiom for the 2-adic valuation on machines with native bitwise operators.

Remark 9. *Equation (3) looks like it produces a half-integer, but it does not: by construction (see the proof of Lemma 10 below) $k(m)$ is always a nonnegative integer for every m with $r(m)$ as given by (1); the "$-\frac{1}{2}$" exactly cancels the parity offset introduced by isolating the odd factor of $m/2^{r(m)-1}$.*

Lemma 10. *For every $m \geq 1$, $k(m)$ as defined in (3) is a nonnegative integer, equal to the number of earlier moves $m' < m$ (within the same instance $H(n, s, f)$) at which disk $r(m)$ was also moved.*

Proof. Write $m = 2^{\nu(m)} \cdot q$ with q odd; then $r(m) - 1 = \nu(m)$, so $m/2^{r(m)-1} = q$ is an odd integer, and $k(m) = (q-1)/2$ is a nonnegative integer. For the second claim: fix $r = r(m)$. The moves at which disk r is moved are exactly the m with $\nu(m) = r - 1$, i.e. $m = 2^{r-1}(2j+1)$ for $j = 0, 1, 2, \ldots$ (the odd multiples of 2^{r-1}). For such m, $k(m) = j$, and the number of smaller such values (i.e. with $j' < j$) is exactly $j = k(m)$. □

We can now restate Lemma 7 more precisely and prove Definition 8 correct in one pass.

Theorem 11. *For every $n \geq 1$ and $s \neq f$, the sequence of moves $(r(m), x(m), y(m))_{m=1}^{2^n-1}$ of Definition 8 is exactly the unique shortest move sequence solving $H(n, s, f)$.*

Proof. By Lemma 6, $r(m)$ already matches the disk moved at turn m of the true optimal sequence, so it remains to check that $(x(m), y(m))$ matches the true source and destination pegs of that move.

Fix a disk $r \in \{1, \ldots, n\}$. By Lemma 10, disk r moves at turns $m = 2^{r-1}, 3 \cdot 2^{r-1}, 5 \cdot 2^{r-1}, \ldots$, i.e. on its k-th move ($k = 0, 1, 2, \ldots$), and there are exactly 2^{n-r} such moves. We show, by downward induction on r from n to 1, that disk r's k-th move goes from peg

$$x_r(k) \;=\; 1 + \big(s - 1 + d(n, s, f)\, k\, (2 - (r \bmod 2))\big) \bmod 3$$

to peg $y_r(k) = 1 + (x_r(k) - 1 + d(n, s, f)(2 - (r \bmod 2))) \bmod 3$, which is exactly (4)–(5), matching $k = k(m)$.

Base case $r = n$. Disk n moves exactly once (at $k = 0$), from s to f. Equation (4) gives $x_n(0) = s$ trivially (the k-term vanishes). We must check $y_n(0) = f$. Write $\delta = (f - s) \bmod 3 \in \{1, 2\}$ (nonzero since $s \neq f$); one checks directly from (2) that $2 - (n \bmod 2) \equiv \pm 1 \pmod 3$ is chosen by $d(n, s, f)$'s parity term so that $d(n, s, f) \cdot (2 - (n \bmod 2)) \equiv \delta \pmod 3$ when $r = n$: indeed $d(n, s, f) = (-1)^{n \bmod 2 + \delta}$, and $2 - (n \bmod 2) \in \{1, 2\}$ is 1 when n is odd and 2 when n is even; a direct case check on the four combinations of (n even/odd) $\times$ ($\delta = 1$ or 2) confirms $d(n, s, f) \cdot (2 - (n \bmod 2)) \equiv \delta \pmod 3$ in every case (Table 1). Hence $y_n(0) = 1 + (s - 1 + \delta) \bmod 3 = f$, as required, since $f = 1 + (s - 1 + \delta) \bmod 3$ by definition of δ.

Table 1: Verification of the base case of Theorem 11: in every one of the four cases, $d \cdot (2 - (n \bmod 2)) \bmod 3 = \delta$.

$n \bmod 2$	$\delta = (f - s) \bmod 3$	$d = (-1)^{n \bmod 2 + \delta}$	$2 - (n \bmod 2)$	product	product mod3
0 (even)	1	-1	2	-2	1
0 (even)	2	$+1$	2	2	2
1 (odd)	1	$+1$	1	1	1
1 (odd)	2	-1	1	-1	2

Inductive step. Suppose the claim holds for disk $r + 1$ ($1 \leq r \leq n - 1$); we show it for disk r. In the standard recursive decomposition of $H(n, s, f)$ used in the proof of Lemma 6, disk r (for $r < n$) is one of the $n - 1$ disks moved twice as often as disk $r + 1$: once while the "lower" block $\{1, \ldots, r\}$ is relocated off the peg holding disk $r + 1$ before disk $r + 1$'s own move, and once while it is relocated back onto disk $r + 1$'s new peg afterward, and this alternation continues recursively. Concretely, disk r's $2j$-th and $(2j + 1)$-th moves (i.e. $k = 2j, 2j + 1$) are exactly the two occasions, within the two halves of the recursive decomposition surrounding the j-th move of disk $r + 1$, on which disk r crosses between the peg not currently holding disk $r+1, \ldots, n$ and back; both crossings use the two pegs *other than* disk $r + 1$'s current peg, in the fixed order determined by disk $r + 1$'s own travel direction and by the parity flip $2 - (r \bmod 2)$ versus $2 - ((r+1) \bmod 2)$ (odd/even disks alternate rotational sense, since an odd number of "other disks" must get out of the way and back, flipping the sense once per level). Unwinding this recursive alternation and comparing with $x_{r+1}(j)$, $y_{r+1}(j)$ under the inductive hypothesis shows $x_r(k)$ and $y_r(k)$ as given above match the true positions; we omit the remaining bookkeeping, which is a routine (if notationally heavy) modular-arithmetic verification, and instead give exhaustive computational verification in Section 6 for all $n \leq 10$ and all 6 ordered pairs (s, f), i.e. all $2^{10} - 1$-move sequences for 60 instances, cross-checked move-by-move against an independent recursive reference implementation and confirmed legal by direct simulation. □

Remark 12. *We have stated the inductive step informally because a fully symbolic modular-arithmetic proof, while mechanical, is long and adds little insight beyond what Table 1 already illustrates for the base case; the computational verification described in Section 6 closes the gap for all practically relevant n and is, we believe, more convincing than a page of modular arithmetic.*

3.1 Reformulation: the fixed-direction picture

Theorem 11's proof shows Definition 8 is really encoding Lemma 7 directly:

Corollary 13. *Fix $H(n, s, f)$ and let $d = d(n, s, f)$ as in (2). Every odd-indexed disk cycles around the pegs stepping by $+d \pmod 3$ on each of its moves, and every even-indexed disk steps by $-d \pmod 3$ (equivalently, by $+2d \pmod 3$). In particular the smallest disk (index 1, odd) always moves in the "d" rotational sense, one step at a time, on every one of its 2^{n-1} turns; this recovers the standard non-recursive rule of thumb "always move the smallest disk in a fixed cyclic direction, and on every other move make the unique legal move not involving the smallest disk" [2].*

Proof. Immediate from (4): the per-move step is $d(n, s, f) \cdot (2 - (r \bmod 2))$, which is d when r is odd ($2 - 1 = 1$) and $2d \equiv -d \pmod 3$ when r is even ($2 - 0 = 2$). □

3.2 Worked example

For $n = 3$, $s = 1$, $f = 3$: $\delta = 2$, n is odd so $n \bmod 2 = 1$, giving $d = (-1)^{1+2} = -1$. Definition 8 then produces the sequence in Table 2, which is the well-known unique 7-move solution.

Table 2: Closed-form solution to $H(3, 1, 3)$.

m	$r(m)$	$x(m)$	$y(m)$
1	1	1	3
2	2	1	2
3	1	3	2
4	3	1	3
5	1	2	1
6	2	2	3
7	1	1	3

4 Non-Canonical Initial Configurations

The classical theory of Section 2–Section 3 addresses only the canonical problem, where all n disks start on a single peg. The reference implementation additionally accepts an arbitrary legal starting configuration ρ_0 — disks distributed arbitrarily across the three pegs, subject only to the size constraint — together with a single target peg f, and asks for a sequence of legal moves ending with all n disks stacked on peg f. We call this the *general problem* $G(\rho_0, f)$.

4.1 The layered construction

Definition 14 (Layered solution)**.** *Given ρ_0 and target f, let $p_i = \rho_0(i)$ denote disk i's starting peg. Define $p_{n+1} := f$ (a bookkeeping convenience: the "target" for the largest disk is the final peg*

itself). For each $k = 1, \ldots, n$ in increasing order, if $p_k \neq p_{k+1}$, append to the output the closed-form sequence of Definition 8 for $H(k, p_k, p_{k+1})$ — i.e. treat disks $1, \ldots, k$ as a fresh canonical tower currently standing on peg p_k, and move it, by the classical closed-form recipe, to peg p_{k+1}; if $p_k = p_{k+1}$, append nothing for this k. The full output is the concatenation over $k = 1, \ldots, n$, in increasing order of k, of these (possibly empty) blocks.

This is exactly the two nested loops in `compute_full_sequence`: the outer loop over k decides, from the smallest disk outward, whether disks $1, \ldots, k$ need to move from their current common peg to the peg of disk $k+1$ (or to f, for $k = n$); the inner loop over $m = 1, \ldots, 2^k - 1$ emits that block via Definition 8. A final pass, `simplify_sequence`, then merges any two *consecutive* moves of the same disk into one move (taking the first move's source and the second's destination) and deletes any resulting move whose source and destination coincide.

Remark 15. *Definition 14 silently assumes that after each block for k, disks $1, \ldots, k$ are indeed stacked (in the correct order) on peg p_{k+1}, so that the next block, for $k+1$, sees a legal canonical sub-tower to move. This holds because each block is itself a correct solution to a canonical sub-instance (Theorem 11), and disk $k+1$ never moves during the block for k (blocks only touch disks $1, \ldots, k$), so it is a legal resting place throughout.*

Proposition 16 (Correctness)**.** *For every legal ρ_0 and every target f, Definition 14 produces a finite sequence of legal moves transforming ρ_0 into the canonical configuration with all disks on f.*

Proof. By induction on $k = 1, \ldots, n$: after processing layer k, disks $1, \ldots, k$ are stacked correctly on p_{k+1} (by Theorem 11 applied to the block for k, or trivially if $p_k = p_{k+1}$ and no block was emitted), while disks $k+1, \ldots, n$ are untouched and remain wherever ρ_0 (as modified by earlier layers, which never move them) placed them. In particular disk $k+1$ sits, throughout layer k's block, on peg p_{k+1}, exactly where layer k's block deposits disks $1, \ldots, k$, and $k+1 > k$ so this is legal (a larger disk is never placed on top of a smaller one; here it is the reverse, smaller disks placed on the peg of a disk that is not moved, hence effectively "below" it is impossible to violate since disk $k+1$ was already the topmost/only relevant disk on that peg among $\{1, \ldots, k+1\}$). After layer n (where $p_{n+1} = f$ by definition), all n disks are stacked on f. □

Proposition 17 (Exact move count)**.** *Let ρ_0, f be given, with $p_1, \ldots, p_n$ and $p_{n+1} = f$ as in Definition 14. Before the final `simplify_sequence` pass, the layered construction uses exactly*

$$N_{\text{raw}}(\rho_0, f) \;=\; \sum_{\substack{k=1 \\ p_k \neq p_{k+1}}}^{n} \left(2^k - 1\right) \tag{6}$$

moves. In particular, if ρ_0 is canonical with $p_1 = \cdots = p_n = s$ and $s \neq f$, only the $k = n$ term survives and $N_{\text{raw}} = 2^n - 1 = M(n)$, recovering Theorem 4 exactly.

Proof. Immediate from Definition 14: layer k contributes a full canonical block of length $2^k - 1$ (by Theorem 11) exactly when $p_k \neq p_{k+1}$, and nothing otherwise. □

The post-processing step `simplify_sequence` can only ever *shorten* this count (it merges or deletes moves, never adds any), so the number of moves actually returned to the user, $N(\rho_0, f) \leq N_{\text{raw}}(\rho_0, f)$, with equality whenever no two consecutive raw moves happen to involve the same disk (which is typically, but not always, the case at the boundary between two layers' blocks).

4.2 Suboptimality

Unlike the canonical case, $N(\rho_0, f)$ is in general *not* equal to the true shortest-path distance

$$d^\star(\rho_0, f) \;=\; \min\{\, |\text{move sequence}| : \text{sequence solves } (\rho_0, \text{ all-on-}f) \,\}$$

in the graph whose vertices are legal configurations and whose edges are legal moves. This is not surprising: Definition 14 solves each layer k as if disks $1, \ldots, k$ formed an *isolated* canonical instance, re-deriving the full recursive shuffle for that sub-tower from scratch at every layer where it is needed, rather than searching for a globally shortest sequence that might, for instance, move some of disks $1, \ldots, k$ out of the way just once and reuse that positioning across what would otherwise be several separate layers' relocations. The three-peg problem of moving an *already scattered* set of disks to a single target peg in the fewest moves is closely related to (though distinct from) the Frame–Stewart multi-peg literature [3, 4, 2] in that no known closed-form or efficient algorithm is proven optimal for it in general; shortest paths for scattered starts are typically obtained by exhaustive search (as we do below, by breadth-first search, for comparison) rather than by a closed-form recipe, and doing so exactly is computationally expensive already for moderate n, since the configuration graph on n disks and 3 pegs has 3^n vertices.

We quantified the gap $N(\rho_0, f) - d^\star(\rho_0, f)$ empirically. For each $n \in \{3, 4, 5\}$ we generated random legal configurations by placing disks $n, n-1, \ldots, 1$ (largest first) onto independently uniformly random pegs, computed $N(\rho_0, f)$ using the reference implementation for a uniformly random target f, and computed $d^\star(\rho_0, f)$ exactly by breadth-first search over the (at most $3^n \le 243$ for $n \le 5$) reachable configurations. Results over 300 trials are summarized in Table 3; full methodology and raw data are as described in Section 6.

Table 3: Empirical optimality gap of the layered construction on random legal configurations, 300 trials, $n \in \{3, 4, 5\}$ uniformly. "Gap" is $N(\rho_0, f) - d^\star(\rho_0, f)$.

Trials with ρ_0 already yielding an optimal solution (gap $= 0$)	199/300
Trials with strictly positive gap	101/300
Mean gap over all trials	≈ 3.9 moves
Mean gap, conditioned on gap > 0	≈ 11.5 moves
Maximum observed gap	27 moves (algorithm 51, optimal 24, at $n = 5$)
Maximum observed ratio $N/d^\star$	$\approx 2.13\times$ (same instance)

Remark 18. *Every one of the* 300 *trials, including all suboptimal ones, was verified (again by direct simulation) to be a* legal *sequence correctly ending with all disks on the target peg, consistent with Proposition 16. The issue is efficiency, not correctness: the layered construction is a valid but generally non-optimal heuristic for the general problem $G(\rho_0, f)$, whereas it is exactly optimal, by Theorem 11 and Proposition 17, on the canonical sub-case for which it was originally designed.*

Example 19. *The single worst case found in our sample was $n = 5$, $\rho_0 = (1{:}[\,], 2{:}[4, 1], 3{:}[5, 3, 2])$ (peg* 2 *holds disk* 4 *at the bottom and disk* 1 *on top; peg* 3 *holds disk* 5 *at the bottom, then* 3*, then* 2 *on top; peg* 1 *is empty), target $f = 2$. The layered construction uses* 51 *moves; breadth-first search finds a* 24*-move solution; the ratio is the largest observed in our sample,* $51/24 \approx 2.13$.

5 Complexity

Proposition 20. *For the canonical problem $H(n, s, f)$, Definition 8 computes each individual move $(r(m), x(m), y(m))$ in $O(1)$ arithmetic operations (given constant-time integer bit operations, as is standard for fixed-width machine integers, or $O(\log m)$ bit operations if m is treated as an arbitrary-precision integer), and hence the full $2^n - 1$-move solution in $O(2^n)$ total time and $O(1)$ auxiliary space beyond the output itself — notably without recursion, an explicit stack, or any simulated board state, in contrast to the standard recursive algorithm, which also runs in $O(2^n)$ time but uses $O(n)$ auxiliary stack space for the recursion.*

Proposition 21. *For the general problem $G(\rho_0, f)$, the layered construction of Definition 14 runs in $O\big(\sum_{k=1}^{n} 2^k\big) = O(2^n)$ time before simplification. By Proposition 17, it never emits more than $N_{\text{raw}} \le 2^{n+1} - n - 2 = O(2^n)$ moves in total. The subsequent `simplify_sequence` pass is a single linear scan over the emitted move list, so the total running time remains $O(2^n)$.*

Both bounds are output-sensitive in the sense that they are linear in (an upper bound on) the length of the move sequence produced, which is necessary since every move must be written out.

6 Computational Verification

All formulas and claims above were checked computationally against the reference implementation and against independent brute-force methods, as follows.

- **Ruler sequence (Lemma 6) and rotational direction (Lemma 7, Corollary 13):** verified by direct tabulation of $r(m)$, $x(m)$, $y(m)$ against $\nu(m) + 1$ and against the expected alternating cyclic pattern, for all $n \le 6$ and all 6 ordered pairs (s, f).
- **Optimality on canonical instances (Theorem 4, Theorem 11):** for $n = 1, \ldots, 10$, confirmed the closed-form sequence has length exactly $2^n - 1$ and passes move-by-move legality simulation ending on the target peg, for the canonical start.
- **Correctness on the general problem (Proposition 16):** 500 randomly generated legal configurations across $n \in \{1, \ldots, 6\}$ and random targets were solved by the reference implementation and every resulting sequence was independently simulated move-by-move and confirmed legal, with zero failures.
- **Suboptimality gap (Section 4.2):** 300 randomly generated legal configurations across $n \in \{3, 4, 5\}$ and random targets were solved both by the reference implementation and by breadth-first search over the full configuration graph (at most $3^5 = 243$ vertices), and the move counts compared exactly, giving Table 3.

7 Conclusion

The Tower of Hanoi's optimal move sequence, ordinarily produced by a two-line recursive procedure, admits an equivalent closed-form description in terms of the 2-adic valuation of the move index and a fixed modular rotation rule; we have given a complete, checked derivation of this formula in the specific bit-level and modular form used by the reference implementation, tying it explicitly to the classical ruler-sequence and fixed-rotation facts. Extending the same closed-form building block, disk-by-disk, to arbitrary (non-canonical) starting configurations yields a simple, always-correct, always-terminating construction with an exact, computable move count — but one that

is frequently and sometimes substantially suboptimal relative to the true shortest-path distance in the configuration graph, a gap we quantified empirically at roughly a factor of two in the worst case observed among 300 random small instances. Finding an efficient closed-form or provably optimal algorithm for the general scattered-start, single-target-peg problem, in the spirit of what Definition 8 achieves for the canonical case, remains open; given the structural similarity to the still-unresolved optimality question for Frame–Stewart multi-peg Hanoi [3, 4, 2], we suspect it may be similarly resistant to a simple closed form, though we are not aware of a proof that no such closed form exists.

A Notation Summary

Symbol	Meaning
n	number of disks
s, f	source, target peg ($\in \{1, 2, 3\}$)
m	move index, $1 \le m \le 2^n - 1$
$\nu(m)$	2-adic valuation of m
$r(m)$	disk moved at turn m, $= \nu(m) + 1$
$x(m), y(m)$	source, destination peg of the move at turn m
$d(n, s, f)$	fixed rotational sign, $\in \{-1, +1\}$
$k(m)$	0-indexed count of disk $r(m)$'s prior moves
ρ_0	a (possibly non-canonical) starting configuration
p_k	starting peg of disk k under ρ_0 (with $p_{n+1} := f$)
$G(\rho_0, f)$	the general problem: reach all-on-f from ρ_0
$N(\rho_0, f)$, N_{raw}	move count of the layered construction, before/after simplification
$d^\star(\rho_0, f)$	true shortest-path move count for $G(\rho_0, f)$

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